Contact

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www.linkedin.com/in/brandon-boesch (LinkedIn)

Top Skills

Place & Route

Embedded Systems

Digital Circuit Design

Languages

English

Honors-Awards

Senior Design Competition Winner (Honors)

Real-Time Operation System's Design Competition Winner

Embedded Systems Design Competition

Engineering Honors

Who's Who Award

Brandon Boesch

Physical Design Engineer at Intel

Austin, Texas, United States

Summary

I earned a Masters and Bachelors degree of Science from the University of Texas in Austin within their Electrical and Computer Engineering program. My primary focus was in Computer Architecture & Embedded Systems with a secondary focus in VLSI hardware design. I am a hard worker, am detail oriented, and strive to be a great leader. My past supervisors will tell you that I am a quick learner, and am always willing to take initiative to get the job done. I enjoy a good challenge and anything that can keep me mentally engaged. When I am not working I enjoy spending quality time with my wife and two kids.

Experience

Intel Corporation

Physical Design Engineer

November 2021 - Present (3 years 8 months)

Austin, Texas, United States

Perform physical design implementation of custom CPU designs starting from RTL and ending at GDS, creating design databases that are ready for manufacturing.

Conduct all aspects of the CPU physical design flow including synthesis, place and route, clock tree synthesis, floor planning, static timing analysis, power/clock distribution, reliability, and power and noise analysis.

Conduct verification and signoff, including formal equivalence verification, static timing analysis, reliability verification, static and dynamic power integrity, layout verification, electrical rule checking, and structural design checking.

Analyze results and make recommendations to improve current and future CPU microarchitectures, closely collaborating with logic, circuit, architecture, and design automation teams.

Possess CPU-specific expertise in various aspects of structural and physical design, including physical clock design, timing closure, coverage analysis, multiple power domain analysis, structured placement, routing, synthesis, and DFT.

Work intimately with industry EDA vendors to build and enhance tool capabilities to design a high-speed, low-power synthesizable CPU.

Optimize CPU design to improve product-level parameters such as power, perfomance, and area (PPA).

Participate in the development and improvement of physical design methodologies and flow automation.

Centaur Technology
Physical Design Engineer
June 2018 - November 2021 (3 years 6 months)
Austin, Texas

Perform back-end physical design based off RTL. After placing and routing a design, perform necessary sign-off checks to ensure the design is both optimal and manufacturable. Required knowledge in all
aspect of IC design, including Synthesis, Static Timing Analysis and Closure, Floor-planning,
Power-grid Insertion, Clock Tree Insertion, Place and Route, Parasitic Extraction, ECO, Signal
EM Analysis, and Dynamic Voltage Drop Analysis.

Assumed partial ownership of the build environment that was
used by our Physical Design Engineers. Ensured that the Build team had the latest and greatest flows available to them. Worked hard to provide to a unified and clean work
environment that was tested and vetted for compatibility across the tools we used.

Arm
IoT Applications Engineer Intern
May 2017 - August 2017 (4 months)
Austin, Texas

First in line resource on help desk for users needing help debugging issues related to hardware and software available within the Mbed OS ecosystem.

Help maintain and improve continuous integration tests for licensed vendors requesting Mbed certification on their devices. Other responsibilities included:
• Pre-release testing of products related to Mbed OS and Cloud connectivity.
• Writing guides and blogs for commonly asked questions. Below is a link to an example I wrote showing users how to recover corrupted boards.
• Showcasing the latest hardware and software by means of sample projects that users can experiment with. Below is an example video and documentation I wrote, showing off the available technology.

Arm
Verification Engineer Intern
May 2016 - August 2016 (4 months)
Austin, Texas

Verified multiple IP elements for the Systems and Software Team. Customized scripts to create a more general test environment, allowing for broader characterizations. Ran regression tests to verify all changes that were made. Used a waveform viewer to ensure protocols were met during interactions between IP elements.

National Instruments
Digital Hardware Engineer Intern
May 2013 - August 2015 (2 years 4 months)
Austin, Texas

I have worked three consecutive summer internships at National Instruments and have had a variety of projects. Primarily focused on board level circuit design and verification.

U.S. Army
Combat Engineer
January 2005 - October 2011 (6 years 10 months)

Performed engineer’s duties in combat and field training environments. Supervised team of 5-10 soldiers and assigned daily duties. Reviewed team’s work to ensure highest standards were met.

Education

The University of Texas at Austin
Masters of Science in Electrical/Computer Engineering, Computer Architecture & Embedded Systems · (2016 - 2018)

The University of Texas at Austin

Bachelors of Science in Electrical/Computer Engineering, Electronics and Integrated Circuits · (2012 - 2016)